

Day 15 – Advanced VPC Networking

Introduction

Amazon VPC allows secure isolated networking inside AWS cloud.

Core Components

VPC, Public Subnet, Private Subnet, NAT Gateway, Route Tables.

Public Subnet

Internet-accessible subnet using Internet Gateway.

Private Subnet

Secure subnet without direct internet access.

Internet Gateway

Enables internet connectivity for public subnet.

NAT Gateway

Provides outbound internet access for private subnet.

Route Tables

Controls routing between subnets and gateways.

Project

Secure multi-tier architecture using public and private subnets.

Security

Security Groups, NACLs, subnet isolation.

Monitoring

CloudWatch monitors network traffic and resource health.

Use Cases

Enterprise apps, banking systems, SaaS platforms.

Hands-On

Create VPC, configure subnets, gateways, and route tables.

VPC Networking Workflow



Networking Components Table

Component	Purpose
VPC	Private cloud network
Public Subnet	Internet accessible resources
Private Subnet	Secure internal resources
NAT Gateway	Outbound internet access
Route Tables	Controls traffic routing

Detailed Workflow

Internet ↓ Internet Gateway ↓ Public Subnet ↓ NAT Gateway ↓ Private Subnet

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